The Manity Health Platform

The Manity Health Platform is an advanced, open, and standards-based EHR and MDM solution tailored for dynamic human operating environments such as space, military, and deep-sea applications. Built on the state-of-the-art FHIR architecture, Manity offers whole-person, longitudinal health data management and seamless vehicle-agnostic integration across diverse operational settings, maintaining interoperability with existing healthcare infrastructure. The platform is designed for complex environments, featuring human factors UI, real-time data ingestion, and advanced CDS hooks for predictive DSIs using evidence-based and AI/ML models. Manity employs the latest HL7 FHIR R5 (v5.0.0) standard to support TRISH HERMES' vision of a semi-autonomous software platform for intelligently managing biomedical, research, environmental, and mission data. Deployed on-premises in a decentralized manner, Manity instances autonomously synchronize via libp2p, enabling automatic discovery and selective synchronization while adhering to data segmentation rules, ensuring redundancy and resilience.

Capabilities Overview

Purpose-Built User Interfaces and Human Factors Design

A key differentiator of Manity versus other legacy EHR systems is our unique approach to platform development. Given the rapidly evolving landscape and dynamic environments inherent in Manity's target client profile, Manity utilizes a standardized back end FHIR architecture paired with one or more modular, bespoke user interfaces (UIs) purpose-built to optimize user, environmental, and mission considerations. Manity supports the SMART on FHIR (Substitutable Medical Applications, Reusable Technologies on Fast Healthcare Interoperability Resources) framework to enable App Store-like functionality within the Manity platform where the user can launch one or more apps that extend the core functionality of the platform. This provides Manity the unique capability to be extensible with 3rd-party provided applications while fully preserving security, privacy, and access scopes.

Security, Privacy, Consent, and Data Access Scopes Management

The Manity platform implements a comprehensive security and privacy model within the core architecture. The model is enhanced via privacy-by-design principles, data segmentation and pseudonymization, and privacy and consent management utilizing a superset of global privacy jurisdictional requirements.

Security

Manity prioritizes security at every level to ensure that sensitive subject and mission data is protected from unauthorized access, tampering, and breaches. To achieve this, multiple security measures are utilized, including secure authentication and authorization mechanisms, encryption of data in transit and at rest, and fine-grained access control policies.

Ingested data is fully encrypted end-to-end, ensuring that information remains confidential and secure during transit between nodes or systems, as well as when stored at rest. The use of modern encryption standards and cryptographic algorithms guarantees that the data is protected from unauthorized access or interception. Additionally, the Identity Provider (IdP) and access control mechanisms ensure that only authorized users can access specific data or perform certain actions on the platform.

Privacy

Manity's architecture prioritizes privacy by employing a comprehensive privacy management approach that incorporates data minimization, pseudonymization, strong access control, encryption, and privacy-preserving data access and sharing methods. The platform is designed with privacy-by-design principles and adheres to global privacy laws such as HIPAA and GDPR. Data minimization is achieved by collecting only the necessary data, while pseudonymization masks personally identifiable information to protect user privacy. Strong access control mechanisms ensure secure and permissions-based data access, while end-to-end encryption secures data both in transit and at rest. For researchers and other stakeholders, the system provides anonymized or aggregated data, preserving user privacy while enabling valuable insights and collaboration.

Consent Management

Manity implements consent management directly into the core architecture via a consent management API built within the identity provider service (IdP). As access requests are received via the IdP, the service evaluates the requests against an extensible consent ruleset that considers research context, data jurisdiction, and user privacy preferences. Please reference the *Architecture > Consent* section below for additional detail.

Data Access Scopes

Manity manages access scopes by implementing a robust role-based access control system that defines different user roles, such as patients, researchers, flight control, in-flight medical operations, crew medical officers, and algorithmic users. Each role is assigned a specific set of permissions and data access levels, ensuring that users can only access the information and functionalities relevant to their roles. This approach maintains data privacy and security while enabling seamless collaboration and data sharing among various stakeholders involved in space missions and healthcare operations. The access control system is integrated with the Identity Provider (IdP) to ensure secure authentication and authorization, enabling a permissions-based data management environment that adheres to regulatory requirements and privacy standards.

Decentralization, Semi-Autonomous Synchronization, and Data Integrity Management

Manity achieves a seamless semi-autonomous longitudinal health record with subject data moving from vehicle to vehicle through a combination of peer-to-peer decentralization protocols (libp2p, mDNS, Gossipsub) combined with data integrity management via vector clocks, and CRDTs (Conflict-free Replicated Data Types).

By using CRDTs, Manity ensures that the subject's data remains consistent across all instances, even in the face of network partitions, latency, or other communication challenges. Manity's' FHIR resources are adapted to accommodate the necessary metadata for vector clocks and CRDTs. Each FHIR resource is extended to include metadata fields to store vector clock timestamps and any additional information required by the CRDT.

Manity supports manual USB or similar physical media patient data synchronization in the event of long-term loss of communication links or systems failures via the FHIR Patient Export operation or Bulk Data Export operations. These operations fully uphold user authorization and access scope security requirements.

Please see additional detail in *Section B, Question 3*.

CDS Hooks and Predictive Decision Support Interventions (DSIs)

Manity ensures agnostic compatibility with external evidence-based, translational research, and predictive (AI/ML) knowledge resources through CDS (Clinical Decision Support) Hooks. This approach allows Manity to exchange clinical and environmental context data with external knowledge sources using a standardized API based on specific events or user actions, enabling context-aware recommendations and interventions within various workflows. By subscribing to forthcoming Clinical Decision Support and predictive Decision Support Intervention (DSI) knowledge sources, Manity can evolve alongside the understanding of the operational and biomedical domain. In the context of HERMES, predictive DSIs can be applied to unique spaceflight use cases, such as monitoring and predicting space motion sickness, detecting early signs of decompression sickness, managing psychological stressors, optimizing sleep and circadian rhythm management, assessing astronauts' hydration and nutrition status, and identifying potential radiation exposure risks.

Research, IRB, and Informed Consent

Manity addresses research consent, datasets, requirements, and IRB and Informed Consent processes via a robust consent management layer utilizing the FHIR Consent Resource within the Identity Provider (IdP). The platform includes a UI for patients and research participants that implements fine-grained preference and privacy access controls as detailed in the IdP Consents section below.

The architecture aligns with the HL7 Vulcan workgroup's evolving efforts, which focus on creating and implementing standardized data models and API specifications for research-related processes, including consent and IRB approvals. In the context of IRB and Informed Consent processes, Manity facilitates the capture, storage, and management of necessary documentation, such as PDFs, doctor's notes, and informed consent documents, consistent with industry best practices interoperability across various research and healthcare systems.

Time Synchronization Management and Data Integrity

Manity will maintain time synchronization and data integrity across distributed nodes via the use of vector clocks and CRDTs (Conflict-free Replicated Data Types). Vector clocks are a mechanism for tracking the causal order of events in a distributed system, allowing the system to maintain data consistency even when nodes communicate intermittently. CRDTs are data structures designed to handle concurrent updates and resolve conflicts without the need for centralized coordination, ensuring data integrity and seamless synchronization.

In the context of a spacefaring mission involving multiple spacecraft and a ground control station, vector clocks and CRDTs play a critical role in managing patient health data and other mission-specific information. For instance, when a spacecraft records a new patient health data point, it updates its vector clock. As the spacecraft communicates with other nodes, they exchange data and vector clocks, ensuring consistent data across all instances while maintaining the causal order of events.

Data Tagging, Domain-Specific Data, and Data Segmentation

Manity utilizes custom FHIR extensions and tags to capture metadata related to patient-attributable and generalized data across different spacecraft and operational environments, providing additional context for medical, biomedical research, environmental, and mission-related data. This enables more granular and flexible data segmentation, particularly in spaceflight relevant research contexts like G-loads, flight types, and day-in-mission. By applying tags to FHIR resources, data is efficiently grouped, categorized, and segmented based on factors such as environment type, mission stage, or data source. This approach allows selective synchronization of patient data while keeping mission, research, or environment-specific data segmented to a subset of peer instances, as illustrated in the example of a patient moving between two spacecraft with custom tags like "SpacecraftA", "SpacecraftB", "G-loads", "FlightType", and "DayInMission" added to the relevant FHIR Observation resources.

Data Storage and Backup

To enable secure, reliable, and redundant data storage minimizing the risk of data loss in the case of system failure or disruption, Manity implements the libp2p decentralization protocol. This protocol facilitates efficient, autonomous data synchronization and replication among multiple instances of the system per data segmentation rules, both on-board the spacecraft and on Earth.

Version Control and Data Provenance

For individual datasets, Manity implements FHIR versioning to track changes in subject data. FHIR versioning allows a unique version id to be assigned to each change to any subject data resource, allowing history tracking and rollback as needed. Additionally, Manity implements the FHIR provenance resource to track dataset origin, modification history, and chain of custody. Manity implements version control for the platform using Git source control.

Power Optimization

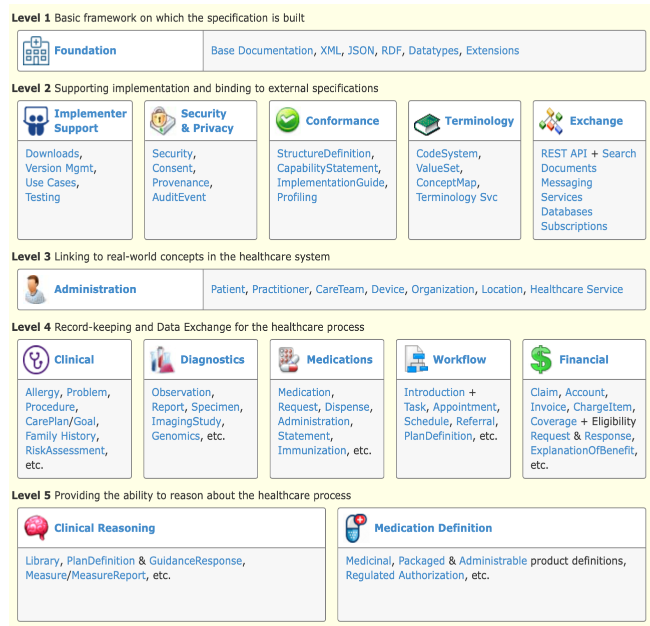
Manity optimizes power consumption in spacecraft and similar environments through selective processing and prioritization on the data ingestion pipeline, processing data based on relevance and urgency. Efficient algorithms, codebase optimization, and data compression techniques reduce energy requirements and the amount of data transmitted between nodes. Adaptive synchronization configures the frequency of peer node communication based on requirements and network conditions, prioritizing critical data and minimizing unnecessary transmissions. By continuously monitoring system power consumption and performance, Manity identifies and addresses inefficiencies and opportunities for optimization, ensuring energy-efficient operation in resource-constrained environments.

Architecture

HL7 FHIR and FHIR Data Model

FHIR is a globally adopted, comprehensive health platform specification that defines a set of capabilities for use across healthcare process, in all jurisdictions, and in many different clinical contexts (https://hl7.org/fhir/modules.html). The Manity Health Platform leverages a broad range of FHIR capabilities to provide architectural advantages as compared to legacy custom EHR data models, HL7 v2/v3 architectures, and general-purpose relational and unstructured solutions. Manity expects and intends to evolve the platform alongside the collective knowledge HL7 FHIR community as well as with requirements informed by the translational and evidence-based space research community.

The Manity data model implements FHIR resources and extensions for storage of biomedical and patient-specific or generalized mission and environmental data. The model includes terminology standardization and resource schema encompassing patient demographics, medical history, diagnoses, medications, vital signs, and many other biomedical, research, device, and observational concepts. FHIR is extensible as needed to accommodate any domain-specific data requirements as well.



FHIR Data Model (https://www.hl7.org/fhir/)

Data Ingestion and Processing Pipeline

To handle the diverse and proprietary non-standard types of data from various medical and diagnostic devices, spacecraft sensors, and a diversity of operating environments, Manity employs a four-step high-performance data management pipeline and privacy-preserving processing engine. The pipeline ensures that data persists at each stage to facilitate processing, validation, storage, and retrieval. Additionally, the architecture allows for flexibility and extensibility, making it easier to accommodate new data sources and formats as they emerge. The data pipeline stages are as follows:

1. **Data ingestion:**
   In this stage, Manity facilitates a variety of connections to various spacecraft systems and biomedical monitoring devices via appropriate APIs or communication protocols. Data is ingested via an extensible plugin architecture that allows for a diversity of device, sensor, and system communication protocols and data standards. Plugins are developed as needed per interface independently from the other pipeline components.

   Raw data from various devices (e.g., wearables, medical equipment, and imaging systems, operating environment) is collected and sent to the system. Example plugins may accommodate HL7 FHIR JSON or XML, HL7 v2.x, x12, NCPDP, CCDA, sensor time-series at various frequencies, and raw video, over a variety of transport protocols. Raw data is temporarily stored in a buffering area utilizing encrypted file or SQL database storage, depending on the data type and volume.
2. **Data normalization, mapping, and transformation:**
   The Manity data transformation engine processes the raw data, normalizing it and converting it into FHIR resources using standard or custom transformation scripts or rules. Manity maintains a data mapping repository to map various device, health, and operational data formats to FHIR resources. For example, one transformation that may occur is a CCDA document type, common in many healthcare settings, to FHIR format.
3. **Data validation, identity, and consent association:**

In this stage, the transformed data is validated against FHIR profiles, ensuring compliance with the FHIR standard. The transformed data is associated with known applicable subject identity(s) contained within the Manity identity server. Data processing consents maintained within the Manity identity server are validated to ensure processing and persistent storage is allowed. Invalid data is logged via an error handling system, while valid data proceeds to the next stage.

4. **FHIR resource storage (processed data):**
   Validated FHIR resources are stored in the FHIR repository. The FHIR resources in this repository are indexed, searchable, and accessible by authorized users via RESTful FHIR APIs.

Management of Diverse Data Sources

The 4-stage data pipeline efficiently handles diverse data ingestion by preprocessing each type of incoming data, such as point measurements, continuous measurements, imaging data, video data, audio data, environmental data, bio samples, omics data, metadata, and unstructured or flat data formats, transforming it into a format compatible with the FHIR standards. To manage various data types and frequencies, the pipeline employs scalable and adaptable data ingestion mechanisms, using buffering and batch processing techniques for continuous measurements and high-frequency data streams, and compression algorithms and encoding techniques for imaging, video, and audio data. Robust data validation, error handling mechanisms, and metadata storage as FHIR Provenance and/or Device resources maintain data integrity and quality throughout the ingestion process, allowing the architecture to seamlessly integrate diverse data for space missions and healthcare operations.

Identity Provider Service (IdP)

Manity implements an identity provider (IdP) service that implements OpenID Connect and OAuth 2.0 to manage identities, credentials, access scopes, authentication, and SSO for the platform. The IdP defines user roles and permissions based on the principle of least privilege, ensuring that users have access only to the data and functionality necessary for their specific role (e.g., subjects, healthcare providers, mission medical personnel). Additionally, the IdP enables the SMART on FHIR protocol and pseudonymization via segmentation of PII.

Consent Management

Manity incorporates a consent management layer within the Identity Provider (IdP) by utilizing the FHIR Consent Resource for capturing essential consent information in a standardized manner. The IdP oversees the entire consent lifecycle, ensuring secure access control and proper authentication. When access requests are made via the FHIR API, the IdP enforces fine-grained access controls in compliance with jurisdictional privacy requirements and consents. Manity also provides a user-friendly interface for patients and research participants to manage their consents, offering an overview, granting or revoking options, and explanations of consent purpose and scope. Consent-related activities are audited and logged for transparency, while interoperability with other FHIR-compatible systems is maintained through the FHIR Consent Resource.

Terminology & Audit / Event Logging Services

Manity utilizes an extensible terminology service to manage code and value sets that standardize the values used in messages between systems. The FHIR standard defines appropriate terminology libraries such as SNOMED and LOINC to describe medical concepts.

Manity utilizes an audit and event logging service to capture all RESTful FHIR operations as FHIR AuditEvent resources. Additionally, the audit and event logging service captures consent-related activities, such as granting, modifying, or revoking consents.

RESTful FHIR API

Manity implements a RESTful FHIR API with access control via the Identity Provider (IdP) service to enable secure, granular, permission-based access to the database for third-party applications. By exposing endpoints for each FHIR resource, applications can perform CRUD operations on the data in a structured and consistent manner, providing them with visualization and analytics capabilities.

The IdP ensures security and access control by handling user authentication and authorization using OAuth 2.0 and OpenID Connect protocols. This token-based approach allows fine-grained access control, enabling third-party applications to access only the specific data and resources they are authorized for, based on the user's permissions. This secure, permission-based access ensures that sensitive healthcare data remains protected while still enabling third-party applications to leverage the data for value-added services.

Software & Hardware Requirements

Operating System: Linux Ubuntu	**Backend Framework**: .NET
FHIR Server: Firely Server	**Frontend Framework**: React
Database: SQL Server 2022 **Web Server**: Nginx	**Authentication & Authorization**: OAuth 2.0, OpenID Connect
Data Processing & Stream Management: Apache Kafka, Apache Flink, and Apache NiFi as needed per mission requirements.	**Monitoring & Alerting**: Prometheus and Grafana **Deployment & Orchestration**: Docker and Kubernetes
Data Synchronization & Peer-to-Peer Communication: libp2p and CRDT-based libraries	**Single Board Computers (SBCs)**: Intel NUC or similar non-proprietary SBCs
Networking: Thunderbolt **Storage**: Utilize SSDs for high-performance storage of frequently accessed data, and HDDs or cloud-based storage solutions for long-term archival of infrequently accessed data.	**Sensors & Data Acquisition**: Manity expects a diverse set of device and sensor requirements as part of the HERMES vision. Specific sensor and data acquisition requirements are expected to be defined as mission parameters become available.

Proposed Project Timeline, Milestones, and Deliverables

Q2 2023 (May - June):

- **Milestone 1**: Develop and finalize project requirements, system architecture, and design documentation. Establish development and testing environments.
- **Deliverable 1**: Demonstration of Medical, research, and environmental/mission data aggregation and ingestion, Immediate availability to local human user, Transfer of data from first (source) implementation of HERMES to a second (destination) implementation of HERMES

Q3 2023 (July - September):

- **Milestone 2**: Implement integrations with external data sources. Develop user interfaces for various user roles (patient, researcher, flight control, and in-flight medical operations).
- **Deliverable 2**: Alpha release, featuring basic core functionalities, user interfaces, and preliminary support for HERMES.

Q4 2023 (October - December):

- **Milestone 3**: Implement peer-to-peer data synchronization, Alpha release is feature complete.
- **Deliverable 3**: Beta release, including required platform features, security measures, and initial third-party system integration capabilities.

Q1 2024 (January - March):

- **Milestone 4**: Conduct extensive testing, performance optimization, and bug fixing. Develop comprehensive documentation, training materials, and user guides.
- **Deliverable 4**: Release candidate, a fully functional platform ready for deployment in real-world scenarios.

Q2 2024 (April - May):

- **Milestone 5**: Gather user feedback and refine the platform as needed for flight readiness
- **Deliverable 5**: Demonstration as defined in HERMES 2023 RFI Section II Deliverables.
- **Deliverable 6**: Documentation as defined in HERMES 2023 RFI Section II Deliverables (including data storage plan, data communication protocols, APIs, and data governance models).

Roadmap, Implementation, System Integration, Training and Support

Manity proposes and anticipates a collaborative, user-centric, and mission requirements driven approach in fulfilling the objectives of this solicitation. Please see the Implementation, System Integration, Training and Support and Roadmap subtopics of the Terms and Conditions section of the solicitation for additional discussion.